



Optimizing variety x sowing date for the farm



The trial finds the best combinations of sowing date and variety for the farm chosen. The combinations ensure that the varieties reach anthesis at the optimum date for the location. The trial also assesses the loss in yield due to straying from those dates. This information enables farmers to decide whether the loss is acceptable in the light of other priorities on the farm.

The approach outlined guides trial managers through designing a study suited to their special requirements. It then explains how to conduct the trial and finally interpret the results. Real life trials are used to illustrate likely responses.

Other companion chapters you will find useful include Why do any of your research on-farm for general information on trials, *Constraints to cereal-based rainfed cropping in Mediterranean environments and methods to measure and minimize their effects* for harvesting methodologies, *What is the best cropping sequence for the farm* if yields are declining on-farm, *Optimizing plant population, crop emergence and establishment*, and *Optimizing nitrogen use on the farm*.

Which farms could benefit from these trials?

- Those farms where the planting date to achieve best yield with the present variety is not known. Optimum planting date may not have been tested but just guessed;
- those farms where it is important to know how much yield is lost by missing the optimum planting date. Farms that might commonly plant late because other crops have still to be harvested or the ground cannot be ready on time;
- those farms that may benefit from varieties with longer or shorter duration or where this option has never been thought about or tested;
- those farms in a region that has new or promising varieties but the varieties have not been tested locally or on the farm.

The best variety

Farmers know that choosing the best variety is the first and probably most important step towards high yield on the farm. But what may not be recognized are the characteristics that are needed in that best variety.

What should the farmer look for?

Most importantly, the best variety must exactly fit around the weather constraints of the farm. Different weather constraints or the same constraints at different times will possibly require a different variety for best yield.

The farmer must know what these constraints are, what they do to the crop and when they occur in the season.

Before starting any trials, check through the following questions on constraints.

Step 1. Work out the dates of constraints

Does weather limit season start?

- On what date approximately is it the right temperature to allow seeds to germinate and grow? When does it warm up enough or cool down enough?
- Is the location dependent on opening rains before the land can be worked or before the crop can be sown? Or is sowing usually held up because the ground is too wet?
- Is the weather limitation always the same and what is the approximate date when it is no longer a constraint? Is this the usual sowing date? How much does that date vary between years?

Is weather limiting during the season?

- Are there weather limitations like very overcast conditions, mists and fogs in the period just prior to normal anthesis date? They can result in low grain numbers and poor yield.
- Can temperatures be very low or very high around anthesis leading to low grain numbers? How many years in ten? If anthesis were slightly earlier or later would it miss these conditions?

Weather at normal anthesis date

- Is there always enough water stored in the soil to fill the grain? Or is there a good probability of sufficient rain late in the season? Do current varieties have problems with pinched grain indicating that water for grain filling may be lacking?
- Have there been problems with small grain? How many years in ten? This is often due to excessively high temperatures late in the season that shorten the grain filling period. What are the temperatures at that time?
- If anthesis were earlier would the constraints be avoided?

Diseases or pests

- Are these likely to be prevalent at particular times? When, on what dates? Are certain stages of plant development more sensitive to the constraint? Is the variety normally used resistant to those threats?

Step 2. Time development to fit around the constraints

Once the approximate dates of the constraints to growth and yield have been identified, the next step is to choose a variety that times its development to avoid or minimize the negative effects of those constraints.

The best variety is one that can be started at or after the opening of the season and then time its anthesis date to avoid frosts and too dull or too hot conditions, and finally and often most importantly, to start and complete grain filling when water and temperature are not constraints.

On top of all these timing requirements, the best variety must have resistance to local diseases, high yield potential and good quality grain.

Early and late varieties

Early wheat varieties are those that reach flowering in a relatively short time from planting while late varieties take longer to complete their life cycle.

Early variety = short duration
Late variety = long duration.

How quickly a variety develops through its vegetative stage into flowering and then through kernel filling depends on its inbuilt responses to particular components of the weather.

Three main factors control how quickly a variety develops

1. Temperature. All varieties develop faster and mature earlier as temperature rises, but they can be delayed a little if it is extremely hot during vegetative growth. Most varieties respond similarly to changed temperature.

2. Day length. Almost all varieties develop faster and mature earlier as the number of hours from sunrise to sunset increases, but varieties differ considerably in how much they speed up with longer days. This is called the day length or photoperiod response of the variety. It is essentially a separate response from that due to temperature.

3. Vernalization. Many varieties develop faster if they have some exposure to cool weather as seedlings. Cool weather means less than 6 °C for strong winter varieties right through to less than 16 °C for extreme spring types. This response to cold is called the vernalization response of the variety. Varieties that respond strongly to cooler conditions take a long time to complete their cycle if they do not experience cool weather. Very low temperatures much below 0 °C have no effect on accelerating development.

The very earliest varieties available respond little to changing day length and have for most practical purposes no vernalization response. They only speed up or slow down as mean temperatures change. In general they are suited to late planting because they can complete their life cycle quickly. As they go through their cycle so quickly they have limited time before anthesis to build up a lot of vegetative growth and the associated structures for very high potential yield.

Late varieties might have either or both photoperiod and vernalization response. They are best suited to early planting because they can take a long time to complete their life cycles. They take a long time to get to anthesis so can generate a lot of leaves and tillers and high potential yield. Whether this yield can be realized depends on components of the environment, particularly water and nutrition.

By combining photoperiod and vernalization responses in different measures, plant breeders can produce varieties with almost any length life cycle. Consequently, there should be at least one variety available that is suited to the special conditions of the farm.

The best sowing date

Does yield change with sowing date?

Though there might be no choice on the first sowing date for the season because it is dependent on first rains or other season break factors, there are choices after that time. Researchers and farmers over decades have examined the effects on yield of changing planting dates. Invariably they have identified the optimum planting time for their chosen variety at their place and found that delaying planting beyond the optimum time results in a decline in yield.

As a typical example, work in Upper Egypt and Sudan found grain yields were 19 percent lower when planting was four weeks after the optimum time (5 percent loss in yield per week).

This was because late planting pushed anthesis and grain filling into hotter conditions thereby shortening the grain filling period and resulting in smaller grains.

Many researchers have argued that the correct sowing time for a variety is the date that gets the crop to anthesis at the optimum time. In the above-mentioned case, this would be before hot conditions constrain yield.

As different varieties take different times between sowing and anthesis the planting date required to reach optimum anthesis date on time must differ between varieties.

A new variety should be calibrated to its new area to ensure that it is sown on the date that gets it to anthesis at the optimum date for the area.

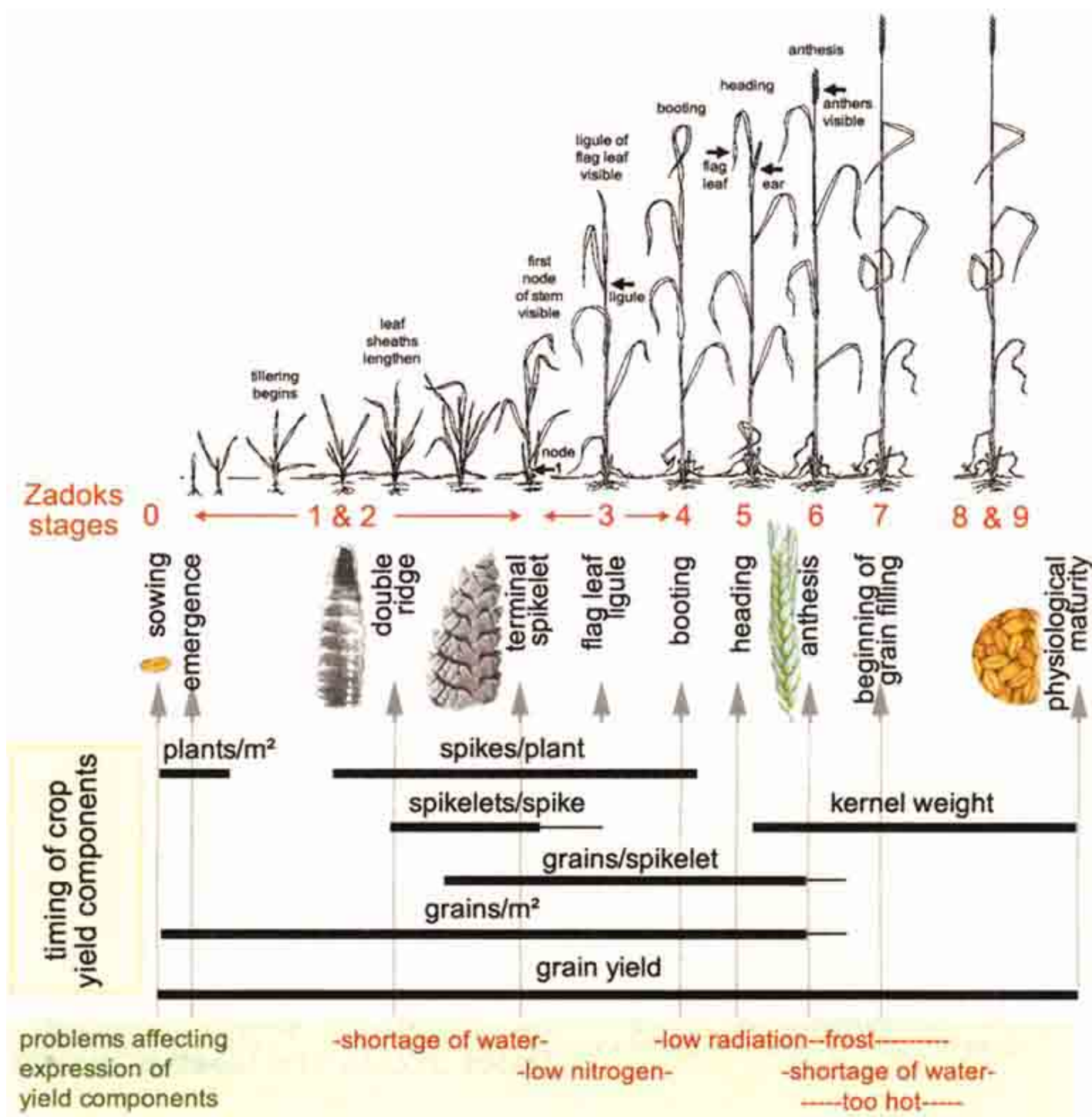
Does yield change with anthesis date?

Just as yield can change with sowing date, yield changes with anthesis date. In an area in southern Australia where yield is high, Stapper and Fischer (1990) found that anthesis date was optimum over only two weeks. Reaching anthesis prior to that period could reduce yield to half that because of frost damage. Reaching anthesis after the optimum period reduced yield by more than 5 percent for every week's delay.

This 5 percent reduction with delay was kept small because they could irrigate. By contrast, missing the optimum anthesis date might be disastrous in areas depending on stored soil moisture or spasmodic though dependable rain to fill the grain. The same would apply if temperatures and rates of evaporation were rising rapidly with advance of the season.

So, for low-yielding Australian crops, the reduction in yield due to delaying anthesis beyond the optimum date was *24 percent per week's delay in a dry year* and from 13 to 9 percent per week's delay in a less demanding wheat zone.

In the chapter on optimizing N a delay of one to eight days only, reduced yield progressively from 3.4 to 2.1 t ha⁻¹ or 38 percent.



Make opportunities to discuss when yield is formed in the crop life cycle.

The progression is plants/m², spikes per plant overlapping with spikelets per spike, grains per spikelet and finally the size of each grain.

Talk about how weather will affect yield at particular stages of development.

The figure will help with the discussion. It shows when parts are developing in relation to how the crop looks. It also shows (in red on green) the danger periods for some types of stress.

For example, if the variety grown on-farm is exposed to frost during the critical period around anthesis when grains (kernels) are just forming, it will yield badly.

That variety should be replaced with one that reaches anthesis well after there is a likelihood of frost, or the sowing date should be adjusted

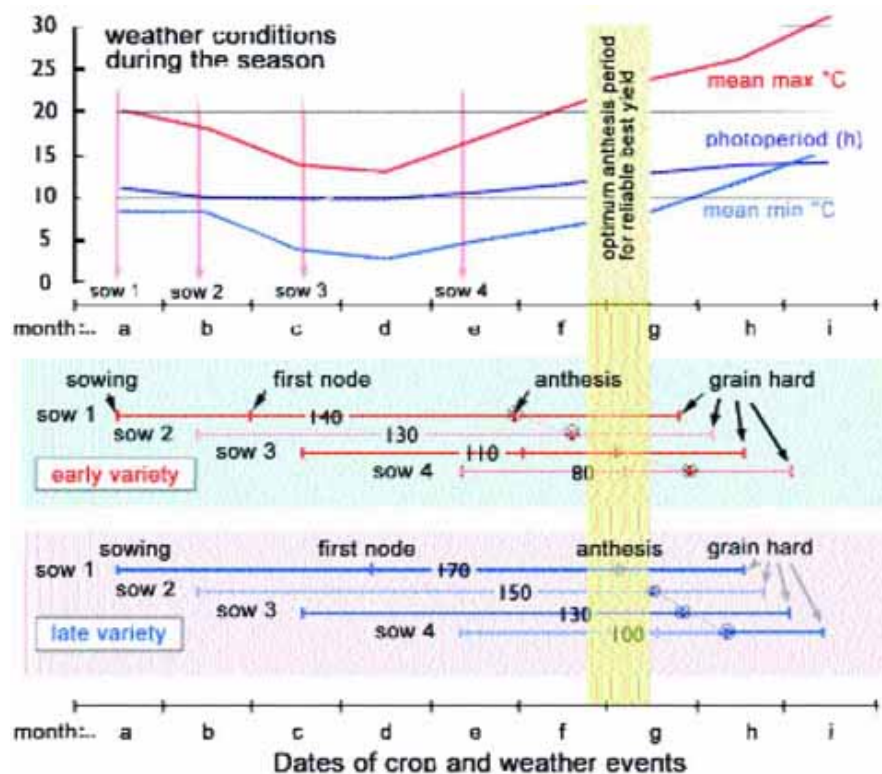
How are sowing date and anthesis date related? A model trial

Farmers know they control the sowing date but may believe they cannot change anthesis date. The following practical but extreme example (Stapper and Fischer, 1990) shows how anthesis date may be changed. Anthesis date can be altered by two means, by changing the date of sowing to a period of different weather and by changing the variety or species to one that develops faster or slower.

In this trial there were two varieties. One was early (i.e. fast) and one was late (i.e. slow), needing long days (long photoperiod) to develop quickly. Of the four sowings, one to three were made over progressively declining temperature and the last sowing (sow 4) during rising temperature. (To see this check the positions of the red vertical arrows on the weather figure immediately below. The names of months are irrelevant.)

- It took plants from the first sowing (sow 1) almost twice as long to reach anthesis as those from the last sowing (sow 4). To see this compare the numbers on the development bars on the figure and the lengths of the bars; e.g. 140d versus 80d. This is primarily because photoperiod was longer throughout growth of late-sown plants and temperatures were rising (see the temperature and photoperiod curves).
- The late variety was always later to reach anthesis than the early one from equivalent sowing dates. The difference ranged from 30 days from sow 1 to 20 days from sow 4.
- Though sowing dates spread over four months, anthesis dates spread over less than two and a half months.

In short, as it gets warmer and photoperiod lengthens, everything goes faster and varieties become more similar in real time. Despite this, variety and sowing date still give a great deal of flexibility when aiming for a particular anthesis date. In this example with extreme varieties, the optimum period of anthesis for the location, shown as a yellow band in the figure, could be targeted by sowing the early variety 100 to 120 days before that period and equally well by sowing the late variety 150 to 170 days before the period. So there was a ten-week window for sowing date to hit the optimum anthesis date by using these extreme varieties. Many varieties are available with rates of development being intermediate between the extreme varieties described, so that a variety could be selected to reach the optimum anthesis period from a wide range of sowing dates.



Designing and doing your trial

General aims and approaches

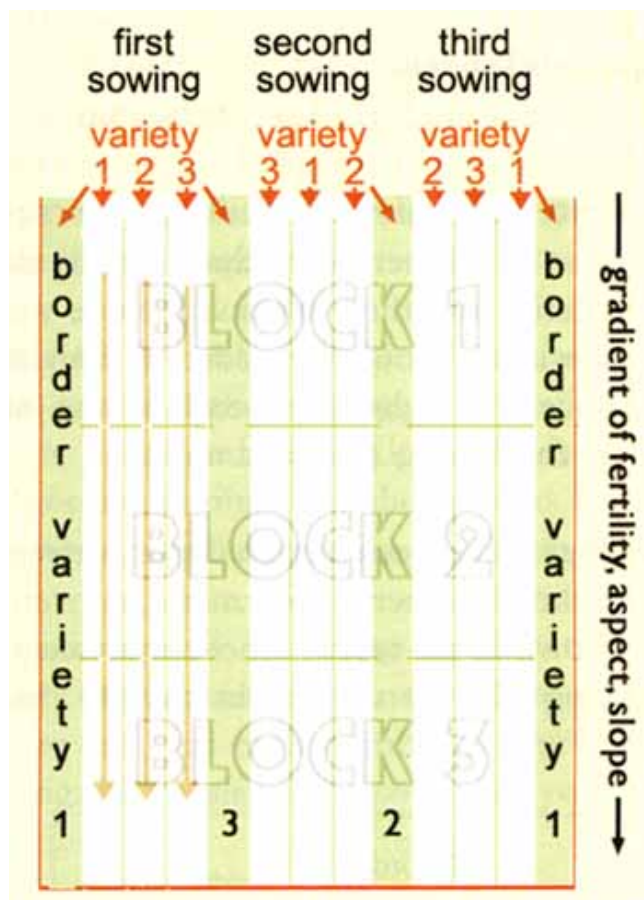
By using varieties contrasting for life cycle length (early, normal and late) and planting at times that span the normal planting date, the trial aims to check whether there are better combinations of variety and sowing date than used traditionally on the farm.

Primarily it aims to identify the optimum date of anthesis. The trial also provides an opportunity to try out any promising new varieties for fitting to the farm's constraints.

Designing for the constraints of the farm

The design shown here as an example has three varieties (early, normal and late) sown on three dates (early, normal and late) and replicated three times (i.e. across three blocks) making a total of 27 plots. For a farmer who is not used to experimental approaches, that size should be just manageable. It is probably also the smallest that will be useful.

A plot layout for determining the best combination of sowing date and variety for the farm. Variety 2 and second sowing are those normally used on the farm.



A realistic number of treatments

On a farm with more resources and where similar studies have been carried out before, the trial can be increased to four (or more) varieties by five dates by three blocks (60 plots). The trial must go right through to completion so must be a size that is both manageable and economically feasible. That said, a study with more varieties and planting dates will identify the best combinations more precisely.

The minimum trial should have 3 varieties (early, normal and late) sown on 3 dates (early, normal and late) and replicated 3 times.

Plot area

Regardless of the number of treatments, each plot should be large enough to allow for a harvest at maturity of at least a 2 m² area surrounded by an uncut border 0.5 m wide. A bigger harvest area with a border is preferable and will actually be simpler to harvest for a farmer who has access to a combined harvester. In any event, the smallest area for harvest should not be less than 1 m². Plots are shown in the diagram as the smallest rectangles edged by a green line. There are 27 of them. They are not drawn to any scale.

Direction of drilling the crop

If the example design is being followed and if the crop is being drilled, each variety is sown lengthwise in a run from the start of the first block right through to the end of the third block (three plots). The three varieties are sown in adjacent strips.

At the first sowing drill the first outside border using the first variety and the last outside border with the last variety. This means there will be two adjacent drill runs of the first and last varieties (see the red arrows on the diagram). If this is a problem, perhaps because of shortage of seed, use the normal variety for all borders.

Follow the general approach at the second and third dates of sowing, except that the varieties are sown in a different order.

Preferably plant the trial within a normal crop as small studies, particularly in a fallow field, can produce unrealistic results because of 'oasis effects'. The blocks can be separated from each other and planted in different parts of the field if it is necessary to avoid small areas that are uncharacteristic or useless for cropping.

Choosing varieties and sowing dates

For the three varieties used in this trial, choose a very early variety (with no day length or cold response), the normal variety for the farm and a late variety. Decide from local considerations whether the late variety should be day length or cold responsive.

For sowing dates choose the normal one for the farm and have one about ten days before and another one ten days after that date. Decide whether a 20-day span is too short or too long to be useful and adjust the time accordingly.

If sowing date on the farm is normally immediately following opening rains i.e. as soon as the main limitation for the area disappears, make that date the trial's first sowing. You might also consider replacing the late variety with a second choice early one to take advantage of having two sowing dates after the farms' normal date.

Preparing the land and sowing

Plots for each sowing date must be cultivated in exactly the same way, each at a similar period in advance of its sowing. All land for the trial should not be cultivated at one time. Cultivating at the one time might disadvantage later sowings because turning the soil accelerates denitrification and more importantly wastes stored water. This complicates the study because of the need to get machinery to plots past other plots.

Apply fertilizer with the seed in each case and if top dressing with fertilizer, this should be done at an equivalent crop stage in each sowing treatment, not at an equivalent time after sowing.

For consistency the same amount should be used in each treatment even though this may not seem sensible.

SEEDLINGS PER SQUARE METRE	Experiment name: <i>variety x sowing date</i>								Recorded by: <i>Jock Mc Tavish</i>			
	Sowing 1 (S1)				Sowing 2 (S2)				Sowing 3 (S3)			
Variety (v)	v1	v2	v3	av	v1	v2	v3	av	v1	v2	v3	av
Block 1	206	189	128	174								
Block 2	222	250	156	209								
Block 3	156	172	133	154								
average	194	204	139	179								

variety names v1: early sowing dates S1: early counting dates S1: Oct 24

v2: normal S2: normal S2:

v3: late S3: late S3:

Observations to make during the trial

- **Weather readings.** These should be taken on a regular basis, preferably on the same day each week, not just when something extreme occurs (see introductory chapters and the example worksheet later in this chapter). Also check for availability of a maximum-minimum thermometer and a rain gauge.
- **Crop observations during the season.** The critical observation to record is the date of anthesis then the date of maturity. Counts of seedlings emerged are also important as variation in numbers between treatments may complicate interpretation of the trial.
- **Weed infestations, diseases and pests** should be noted and controlled as in a normal crop. If possible a record should be kept of their severity for all plots using a rough score out of ten. Different yields between treatments may be due to factors like these that are not intended as part of the trial and differ between seasons. Such observations help in a final assessment.

Harvest methods for harvesting trials at grain maturity are in the chapter *Constraints to cereal-based rainfed cropping in Mediterranean environments and methods to measure and minimize their effects*.

Example worksheets

Whenever possible, worksheets within a trial should use a common design. The first worksheet is for entering counts of seedlings per square metre in an experiment with three sowing dates, three varieties and three blocks. If there were a fourth sowing it would be added as four more columns at the right hand end of the table. Numbers for sowing 1 are examples.

If seedlings have been counted per metre row length rather than per square metre, convert the values shown in the table to those equivalents by multiplying by row spacing in cm/100. So for 18 cm drill rows, 222 m⁻² becomes 40

seedlings per metre row. The layout of the table is intended to make comparisons between data sets easy. It can be seen straight away in the example data that emergence was poor in Block 3 and that variety 3 was poor in all blocks.

GRAIN YIELD PER SQUARE METRE	Experiment name: <i>variety x sowing date</i>								Recorded by: <i>Jock Mc Tavish</i>			
	Sowing 1 (S1)				Sowing 2 (S2)				Sowing 3 (S3)			
Variety (v)	v1	v2	v3	av	v1	v2	v3	av	v1	v2	v3	av
Block 1	89	100	189	126								
Block 2	111	133	200	148								
Block 3	100	111	150	120								
average	100	115	180	131								

variety names v1: early sowing dates S1: early counting dates S1: *Jan 20*
v2: normal S2: normal S2: *Jan 26*
v3: late S3: late S3: *Jan 31*

SUMMARY WORKSHEET

sowing date	wheat variety	seedling number (/m ²)	anthesis date	grain yield (g/m ²)	total biomass (g/m ²)	harvest index	sowing - anthesis (days)	sowing - anthesis (°Cd)
1 Sept	v 1	194	1 Nov	100	286	0.35	61	1000
	v2	204	10 Nov	115	245	0.47	70	1200
	v3	139	15 Nov	180	462	0.39	75	1300

The worksheet for grain yield below copies that for seedling emergence.

A table for biomass (weight of grain, straw and leaves) would be the same again as would that for anthesis dates. A familiar design leads to fewer mistakes and writing down the numbers is faster as you get to know the positions of cells on the sheet.

As farmers will be using these worksheets, explain that having a standard layout with a box, or cell, for each number not only makes entering data easier but also makes your job of transferring data onto a computer spreadsheet easier, if this is an option and your preference.

It is always useful to have a summary worksheet so that you can see the relationships between the main variables of the trial. Its data will be extracts from the raw worksheets, usually the averages taken from the bottom rows. It might look something like the above summary table that shows the first sowing.

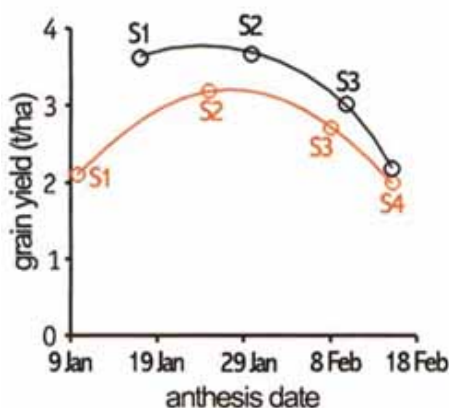
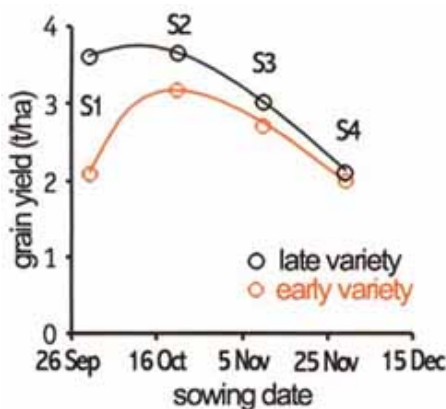
The worksheets for the rainfall and temperature numbers could be set out as below with a cell for recording each variable on a standard day each week. The average temperature (av °C) is worked out as the maximum plus the minimum divided by two. Thermal time summation for each week is the average temperature multiplied by seven (days in the week) and the units are °C d.

Rain, max & min temperature & heat sums each week					Experiment name: <i>Variety x sowing elate</i>								Recorded by: <i>Jock Mc Tavish</i>							
date	18Nov	25 Nov	2 Dec	9 Dec	16 Dec	23 Dec	30 Dec	6 Jan	13 Jan	20 Jan	27 Jan	3 Feb	10 Feb	17 Feb	24 Feb	3 Mar	10 Mar	17 Mar	24 Mar	31 Mar
rain (mm)	5	37	0	0	12	0	0	40	22	700	0	0	0	0	74	22	0	0	0	0
max °C	30	31	30	28	23	25	26	22	20	27	25	25	23	18	23	30	32	29	34	38
min °C	14	13	10	12	9	11	8	10	8	9	9	11	13	6	11	12	14	15	14	18
av °C	22	22	20	20	16	18	17	16	14	18	17	18	18	12	17	21	23	22	24	28
week °Cd	154	154	140	140	112	126	119	112	98	126	119	126	126	84	119	147	161	154	168	196
sum °Cd	154	308	448	588	700	826	945	1057	1155	1281	1400	1526	1652	1736	1855	2002	2163	2317	2485	2681

Once you have the results

Using graphs to calculate optimum sowing and anthesis date and yield loss for being late.

Association between sowing date and grain yield (up) and anthesis date and grain yield (down). The example has two varieties (late and early) and 4 sowing dates (S1-S4)



Once the data are available, the first things collaborating farmers will want to know is what was the best sowing date and the best variety. There will have been an indication of the answers when the plots were harvested, but no detailed idea of the trends.

Commonly the best way to show trends is on a time-based graph. Find the sowing dates, anthesis dates and yield data for each plot on the worksheets and then calculate the average yields and anthesis dates across blocks. Put those numbers on a figure like the left hand one below, scaling it to the appropriate dates and yields.

The conclusion from the example left hand figure is that:

- the late variety is best as it yielded more than the early variety from all sowing dates;
- the best general sowing date is date 2, though the late variety did equally well from date 1;
- sowing dates 3 and 4 are past the peak for both varieties with yield declining at about 270 kg per week delay in sowing (around 7 percent per week).

Interpret the farms' data similarly. If three sowing dates were used the pattern may be less clear than in the example.

Return to the example. Sowing 1 is too early for the early variety. The right hand figure of the same yield data, plotted against anthesis date instead of sowing date, shows why. According to that figure the optimum anthesis date for the area is about 25 January (when the curves for both varieties reach their peaks).

The early variety from sowing 1 reached anthesis two weeks too early for this date and may have lost yield through frost or by having too little biomass accumulated by anthesis to support high yield. Together the curves indicate that the late variety might have yielded best from a sowing date between sowings 1 and 2. See whether there is any indication of an optimum anthesis date amongst varieties for your data by drawing curves through the data for each variety separately. Do the curves peak somewhere near the same date?

The anthesis date figure is very useful for forecasting best sowing dates for varieties that were not included in the trial.

Whatever the variety, they must aim to have anthesis at the optimum time. An earlier variety than used in the example would need to be sown possibly later than sowing 2 to reach anthesis at the optimum time. A really late variety might have to be sown before sowing 1. The common peak anthesis time is what should be aimed for each year unless the weather data collected by farmers indicate that the year of the trial was abnormal.

Using heat sums to calculate latest sowing date once you know optimum anthesis date

The following calculation is approximate but can indicate what is the very latest date that a short duration variety can be sown to reach anthesis at the optimum anthesis date.

Working out sowing date from anthesis date using temperatures collected for the area

These numbers are from the model trial "How are sowing date and anthesis date related?"

	month:	b	c	d	e	f
1	average temperature (°C)	12.9	8.6	7.9	10.6	13.1
2	days in the month	31	30	31	31	30
3	°Cd in the month (1 × 2)	400	258	245	329	393
4	°Cd totalling back from month "f"	1624	1225	967	722	393

What this calculation needs is mean temperatures (average of maximum and minimum) for the area. Farmers will have collected these during the trial.

Crops set their rate of development primarily by their inbuilt temperature clock. Each day during growth this clock adds the current day's average temperature to its running total and when the total reaches particular sums the crop can start its next stage of development.

These totals differ between stages and between varieties. They are referred to as heat sums, thermal time or day degrees and the units are °C d (°C each day accumulated over a number of days). For example, on a day with an average temperature of 20 °C a crop accumulates 20 °C d to be added to its running total.

The very shortest duration wheat varieties need to accumulate about 1200 °C d to develop between sowing and anthesis. That could be 100 days with an average temperature of 12 °C (100 days x 12 °C = 1200 °C d) or 60 days at 20 °C (60 days x 20 °C = 1200 °C d) or some other combination.

To work out the approximate latest sowing date from the optimum anthesis date, successively total the day degrees for the days prior to the optimum anthesis date until the desired total for the variety is reached (1200 °C d in this case). The date at which the total heat sum (1 200 °C d) is reached is the latest sowing date for a short duration wheat. Collaborating farmers will probably have taken temperatures weekly, so will accumulate mean °C x 7 weekly.

The example in the table is based on weather data from the trial on page 4. The optimum anthesis date there was end of month "f".

To reach anthesis at that time a 1 200 °C d variety would need to be sown early in month "c" (in the table see the backward summation of °Cd from month "f" that reaches 1 225 °C d in month "c").

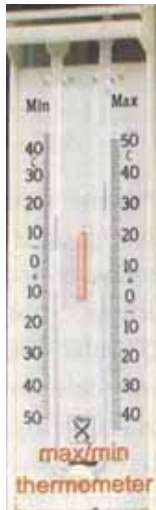
Heat sums may seem complex but they do lead to a better understanding of what makes crops develop. Persevere with this example. You only need temperature data.

Old weather records could save you work

If local farmers or neighbours have been collecting weather information (temperature and rainfall) over the years, or if you have access to long-term weather data for the region, all of the trial may not be needed. You can calculate best sowing and anthesis dates from such data if you know enough about the varieties. You can also assess how often, e.g. how many years in ten, yield is likely to decline because of variations in weather.

The necessary frost-free period

The frost-free period should extend from when the spike starts to emerge preferably until well into grain filling. If temperatures go down to 0 °C for even one night during this time some florets will be sterilized and yield will decline. Several consecutive nights of such temperatures or just one -5 °C could lose much of potential yield.



H.M. RAWSON

You can work out roughly how long this frost-free period needs to be.

Depending on average temperature the crop needs two to four weeks before and two to six weeks after anthesis without frost (warmer equals shorter periods).

Using the long-term weather information, note down the beginning and end of the frost-free period and check if it applies in all years. In how many years is the frost-free period too short? If the frost free period is commonly much longer than the required four to ten weeks, still note down the starting and finishing dates. These observations will give you your frost-free period and its variability in starting and finishing date from year to year.

Amount of water for grain and biomass

Wheat requires at least 200 litres of water during grain filling to produce each 1 kg grain and up to twice that amount in very hot, dry areas.

Scaling that ratio up, a 3 000 kg (3 t ha^{-1}) crop will need 600 000 litres (0.6 ML) either from stored soil moisture or from rain. This is equivalent to 60 mm rain (1 mm rain falling on 1 m^2 is 1 litre or over 1 ha is 10 000 litres).



The crop also needs water to grow prior to anthesis. For a 3 t ha^{-1} yield the crop will need to have more than 6 t ha^{-1} of biomass produced by anthesis, requiring around 160 mm rain or equivalent stored moisture (a clay near field capacity could contain this amount).

If the long-term weather data show this amount of water is not available in most years, yields will be less than 3 t ha^{-1} and a short duration variety might be most efficient for the area.

There is no point producing a lot of biomass before anthesis, as is possible with a long duration variety, if there is no water remaining after anthesis to fill the grains.

Evaporation

If the air is very dry and it is hot, sunny and windy, crops potentially lose a lot of water, up to 10 mm a day. If it is cool and sunny losses are much lower. Under the former conditions crops are inefficient in their use of water, under the latter they can produce more biomass or yield for each mm of water they transpire. Consequently, if water is a critical limitation, it is better to avoid these periods of high evaporation during the anthesis and grain filling period.

In warm areas where frost seldom occurs during the cooler period of the year, this cool period of lower evaporation might be the best time for grain filling. The coolness also extends the period of grain filling.

Working with numbers in a trial

The following trial is entirely imaginary, designed as an introduction to working with numbers and for looking at, thinking about and interpreting a moderately complex set of data that you can discuss with collaborating farmers.

Description of the imaginary field

The example small farm has no uniform areas. The best field slopes from a small hill where the effective soil depth is about 40 cm, down to a shallow hollow with deep soil that can be damp for some time after rain. The hollow is prone to frost in some years and collects mist after cool nights late in the season.

The imaginary trial was planted in a strip to include the hollow (Block 1) the slopes (Block 2) and the exposed hill (Block 3). Due to the variability of the field the blocks could also be thought of as subtrials 1, 2 and 3.

The varieties

The varieties included: a very short duration one (early), not normally used in the region (v1); two local high yielding mid season varieties (one flowering 5-12 days later than the other depending on planting date (v2 and v3); and a long duration (late) variety known to be variable in yield from year to year (v4).

The sowing dates

Of the four sowing dates used (S1 to S4), the third (S3) is considered best for the region; any later sowing (S4) would expose the crop to high temperatures and hot winds during grain filling. S1 is probably too early.

The yield data

The large table for grain yield with its four varieties by four sowings by three blocks will appear very complicated to farmers at first sight. So start your discussion by looking at the following small table. It summarizes the trial by averaging its three blocks.

	v1	v2	v3	v4	variety average
Sowing 1	1.0	1.2	1.5	1.6	1.3
Sowing 2	1.9	2.7	3.5	3.6	2.9
Sowing 3	3.0	3.4	3.7	3.5	3.4
Sowing 4	2.2	1.8	1.3	0.7	1.5
sowings-average	2.0	2.3	2.5	2.3	2.3

These summary data show that the general expectations for the imaginary farming area were supported as follows:

- the best sowing date was sowing 3 as it gave the highest yield on average for the four varieties (last column, 3.4 t ha⁻¹). The first sowing was too early with only 1.3 t ha⁻¹;
- a local variety, the medium late v3, was the best across sowings, yielding 2.5 t ha⁻¹ (bottom row).

However, there were new and interesting findings:

- the long duration variety (v4) did best from the early sowing (1.6 t ha⁻¹), but very badly from the last sowing (0.7 t ha⁻¹);
- the short duration variety (v1) did creditably and by far the best from the final sowing (2.2 t ha⁻¹). This indicated that a late sowing using a short duration variety would be an option if the ground was not ready at the normal date of planting.

The individual data from the blocks (now see the large workbook table) also reveal some interesting trends that tell something about crop responses to different conditions in the blocks. Recall that Block 1 is in a dip in the field, Block 2 is on a slope and Block 3 is at the top of a rise.

Overall, the blocks were very similar in performance (2.2, 2.3 and 2.4 t ha⁻¹; you have to calculate these numbers from the big table) but the patterns across blocks from different sowings were very different. The block differences can be interpreted as follows.

The early sowing: severe frost damage caused the short duration variety to yield very poorly from sowings 1 and 2 in the low-lying area of Block 1 (0.8 and 1.4 t ha⁻¹). On the windy exposed hill of Block 3, frost damage to these plantings was far less pronounced (1.2 and 2.3 t ha⁻¹).

By contrast the long duration variety did better in Block 1 than Block 3 from the early sowing (1.7 and 1.3 t ha⁻¹). This was because it flowered after the frost events could cause damage, but an unrelated problem of high wind caused the exposed crop in Block 3 to lodge slightly, so that it yielded less well.

The last sowing: a contrast between Blocks 1 and 3 also occurred at the last sowing because of the difference in terrain. In this case, Block 1 was superior because, being in a dip with deeper soil, it had more water to take it through the rapidly drying period of grain filling. The late variety, because it filled its grain late in the season, was a disaster from the last sowing in Block 3 because of water shortage (0.3 t ha⁻¹).

Being baffled with too many numbers

The above-mentioned analysis is concocted to illustrate that there can be a lot more information in a data set than shows in averages.

Indeed, farmers might find the host of numbers collected in a trial baffling, just because there are so many. If so, condense the numbers into a few familiar averages, like average grain yield for the whole trial. This will help digest a big data set.

Grain yield converted to t ha⁻¹ for all 4 varieties (v1 early, med early, med late, v4 late) 4 sowings (s1 early, med early, med late, S4 late) and 3 blocks as would be recorded in the workbook. Because of the variability of the field the blocks could have been treated as sub experiments

GRAIN (t/ha)	Experiment name: <i>Imaginary</i>										Recorded by: <i>U. Tom Cobbleigh</i>									
	Sowing 1 (S1)					Sowing 2 (S2)					Sowing 3 (S3)					Sowing 4 (S4)				
Variety (v)	v1	v2	v3	v4	av	v1	v2	v3	v4	av	v1	v2	v3	v4	av	v1	v2	v3	v4	av
Block 1	0.8	0.9	1.3	1.7	1.2	1.4	1.9	3.2	3.2	2.4	3.1	3.1	3.4	3.2	3.2	2.4	2.0	1.6	1.2	1.8
Block 2	1.0	1.2	1.6	1.8	1.4	1.9	2.8	3.8	3.7	3.1	2.9	3.5	3.8	3.7	3.5	2.0	1.8	1.3	0.5	1.4
Block 3	1.2	1.4	1.7	1.3	1.4	2.3	3.5	3.6	3.9	3.3	3.0	3.5	3.9	3.6	3.5	2.2	1.5	0.9	0.3	1.2
Average	1.0	1.2	1.5	1.6	1.3	1.9	2.7	3.5	3.6	2.9	3.0	3.4	3.7	3.5	3.4	2.2	1.8	1.3	0.7	1.5

This can be compared against normal yield last year and so on. You can then work back into the slightly more complex but still condensed data looking at, for example, average yields of different varieties.

Once you have thought and discussed this and its meaning you should drop into the next level of detail until you get right back to the raw data. Explain that bit by bit you can unravel the full story. It is useful, however, to start at the gross level of overall averages.

Working with harvest index

The plot samples in the imaginary trial were dried and then weighed before threshing so those weights can now be used to work out how efficient the crop treatments were in producing grain.

The aim of most cereal farmers is to have as much grain as possible produced for every unit of biomass that grows. The proportion of grain to total above-ground matter (grain, straw and leaves) is called harvest index (HI).

An efficient crop produces around 50 percent of its above-ground dry weight as grain. This can fall below 30 percent (HI of 0.3) in frosted, lodged or droughted crops and lower still if some problem like boron deficiency has sterilized some ears.

In the current trial the usefulness of working out HI is that it will indicate if there were any problems, shown as depressed HI, during the growth of the crop.

The first step is to have the total above-ground dry weight data in a worksheet table like that containing grain weight data. Then make a blank table in which to write down the HI numbers.

HI is calculated by dividing grain weight for the plot by the total harvested weight for the plot. It can also be expressed as a percentage by multiplying the result by 100 as in the accompanying table.

Working out harvest index from grain yield and biomass (t ha⁻¹) for sowings 1 and 3 in Block 1

	(v1)	(v2)	(v3)	(v4)	av
	early			late	

grain yield					
Sowing 1	0.8	0.9	1.3	1.7	1.2
Sowing 3	3.1	3.1	3.4	3.2	3.2
biomass					
Sowing 1	4.7	5.0	6.8	10.6	6.8
Sowing 3	5.7	6.0	6.7	6.7	6.3
HI%					
Sowing 1	17	18	19	16	17
Sowing 3	54	52	51	48	51

The HI table shows grain yield, biomass yield and HI for sowings 1 and 3 of Block 1 in the fictitious trial. The grain yield data have been shown in the earlier large table.

The bottom two values in the last column of the HI table are average harvest index for sowings 1 and 3. They demonstrate that there was very little grain produced for the amount of biomass in sowing 1 compared with sowing 3. As discussed, this was because frosts reduced fertility more than biomass in early and medium early crops of sowing 1 and the medium late and the late crop was lodged.

Values above 50 percent for HI in sowing 3 show that very little went wrong with those crops. They yielded near their potential.

Scan the Diagnostic Key diagram in the chapter *Constraints to cereal-based rainfed cropping in Mediterranean environments and methods to measure and minimize their effects* to see what might cause low HI in wheat crops.

Further reading

Stapper, M. & Fischer, R.A. 1990. Genotype, sowing date and plant spacing influence on high-yielding irrigated wheat in southern New South Wales. III Potential yields and optimum flowering dates. *Australian Journal of Agricultural Research* 41, 1043-1056

